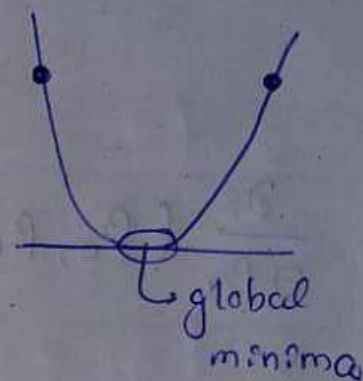
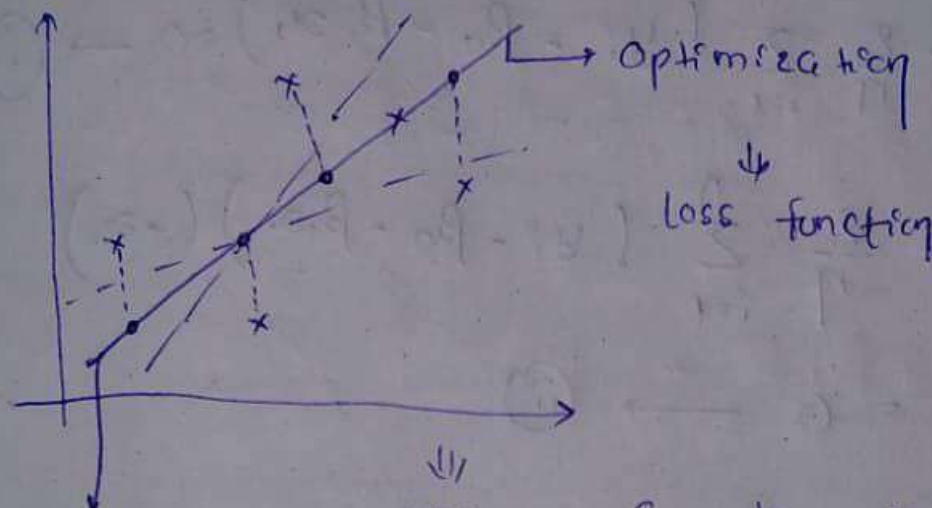


Linear Regression Using OLS

65

Ordinary Least Square



$$h_0(x) = \beta_0 + \beta_1 x$$

$$\quad \quad \quad \hookrightarrow \hat{y}$$

$$\xrightarrow{\text{OLS}} \text{ formula and calculate } \beta_0 \text{ \& } \beta_1$$

Ordinary least square

$$S(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

$$\text{Aim} \quad \underline{\beta_0} \text{ \& } \underline{\beta_1}$$



$$\xrightarrow{\text{Minimize}} \text{MSB}$$

$$\begin{cases} y = x^2 \\ \frac{\partial y}{\partial x} = 2(x)^{2-1} \\ \frac{\partial}{\partial x}(x^2) \end{cases}$$

$$\frac{\partial S}{\partial \beta_0}(\beta_0, \beta_1) = \frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) (0 - 1 - 0) = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \rightarrow \textcircled{i}$$

$$\frac{\partial S}{\partial \beta_1}(\beta_0, \beta_1) = \frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) (-x_i) = 0 \rightarrow \textcircled{ii}$$

$$= -\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) (x_i) = 0 \rightarrow \textcircled{ii}$$

eq $\rightarrow \textcircled{i}$

$$-\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$-\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$-\sum_{i=1}^n y_i + n\beta_0 + \beta_1 \sum_{i=1}^n x_i = 0$$

$$n\beta_0 + \beta_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$$

$$\beta_0 = \frac{\sum_{i=1}^n y_i}{n} - \beta_1 \frac{\sum_{i=1}^n x_i}{n}$$

Intercept $\rightarrow$

$$\boxed{\beta_0 = \bar{y} - \beta_1 \bar{x}}$$

Eq (1)

$$-\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) (x_i) = 0$$

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) (x_i) = 0$$

$\Downarrow$

$$\sum_{i=1}^n x_i y_i - \beta_0 \sum_{i=1}^n (x_i) - \beta_1 \sum_{i=1}^n (x_i)^2 = 0$$

$$\sum_{i=1}^n (x_i y_i - \beta_0 x_i - \beta_1 x_i^2) = 0$$

$\Downarrow$

Replace $\beta_0 = \bar{y} - \beta_1 \bar{x}$

$$\sum_{i=1}^n (x_i y_i - (\bar{y} - \beta_1 \bar{x}) x_i - \beta_1 x_i^2) = 0$$

$$\sum_{i=1}^n (x_i y_i - x_i \bar{y} + \beta_1 \bar{x} x_i - \beta_1 x_i^2) = 0$$

$$\sum_{i=1}^n (y_i - \bar{y} + \beta_1 \bar{x} - \beta_1 x_i) x_i = 0$$

$$\sum_{i=1}^n [(y_i - \bar{y}) + \beta_1 (\bar{x} - x_i)] = 0$$

$$\sum_{i=1}^n (y_i - \bar{y}) + \sum_{i=1}^n \beta_1 (\bar{x} - x_i) = 0$$

$$\sum_{i=1}^n \beta_1 (\bar{x} - x_i) = - \sum_{i=1}^n (y_i - \bar{y})$$

$$\beta_1 = \frac{- \sum_{i=1}^n (y_i - \bar{y})}{\sum_{i=1}^n (\bar{x} - x_i)}$$

$$\beta_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})}$$

$$\beta_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})} \Rightarrow \text{coefficient}$$

OLS

$$\beta_0 = \bar{y} - \beta_1 \bar{x} \Rightarrow \text{Intercept}$$

Intercept
coefficient $\downarrow$

$\downarrow$

$$\beta_0 = (\bar{y} - \beta_1 \bar{x})$$

x	y	$(y_i - \bar{y}) \div (x_i - \bar{x})$	β_1
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-
$\bar{x}$	$\bar{y}$		

OLS $\approx$ Linear Regression (sklearn)